

UFO Sightings

Sistemas de Informação 1

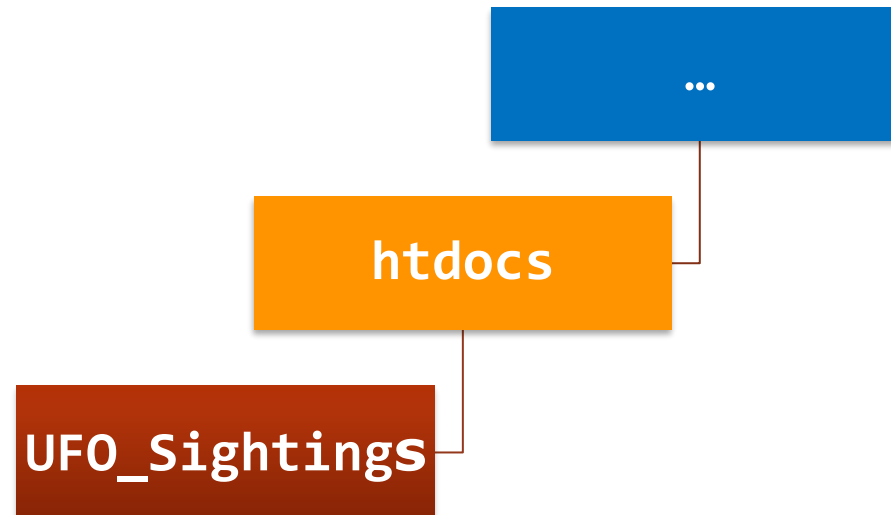
Setup Environment

- On Windows, you can download and install XAMPP.
 - XAMPP is a very easy to install Apache Distribution for Linux, Solaris, Windows, and Mac OS X.
 - The package includes the Apache web server, MySQL, PHP, Perl, a FTP server and phpMyAdmin.
 - Url: <https://sourceforge.net/projects/xampp/>
- On MacOS, you can download and install MAMP
 - MAMP is a free local server environment with Apache, Nginx, PHP, and MySQL.
 - Url: <https://www.mamp.info/en/mac/>



Create the project

- Go to **htdocs** folder
 - This is where the web server expects to find the web site
 - On Windows: “**.../xampp/htdocs**”
 - On MacOs: “**/Application/MAMP/htdocs**”
- Create a folder named **UFO_Sightings**

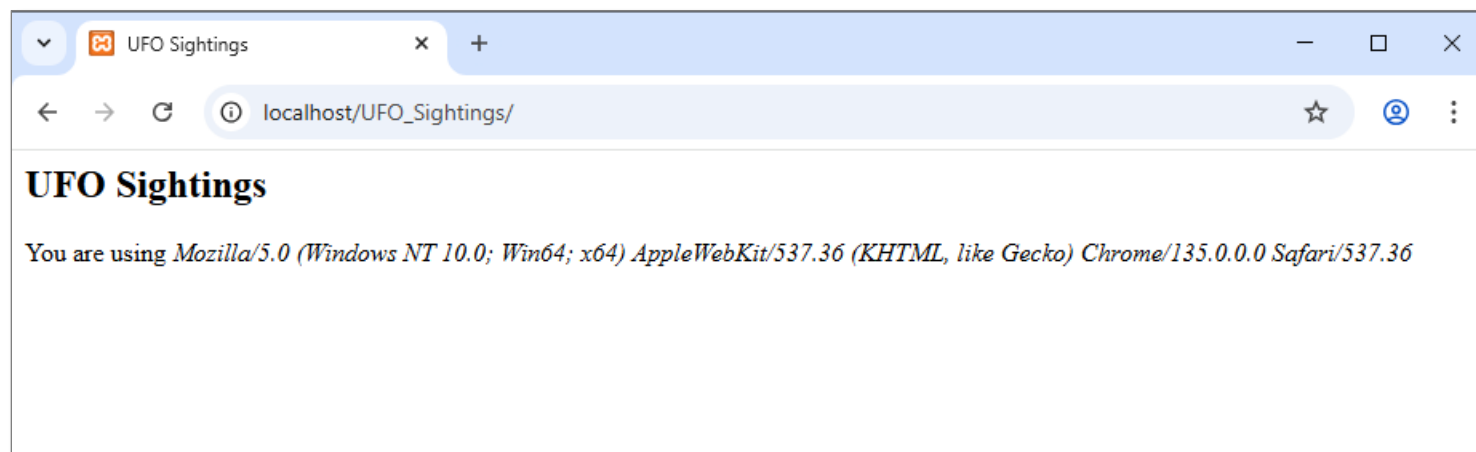
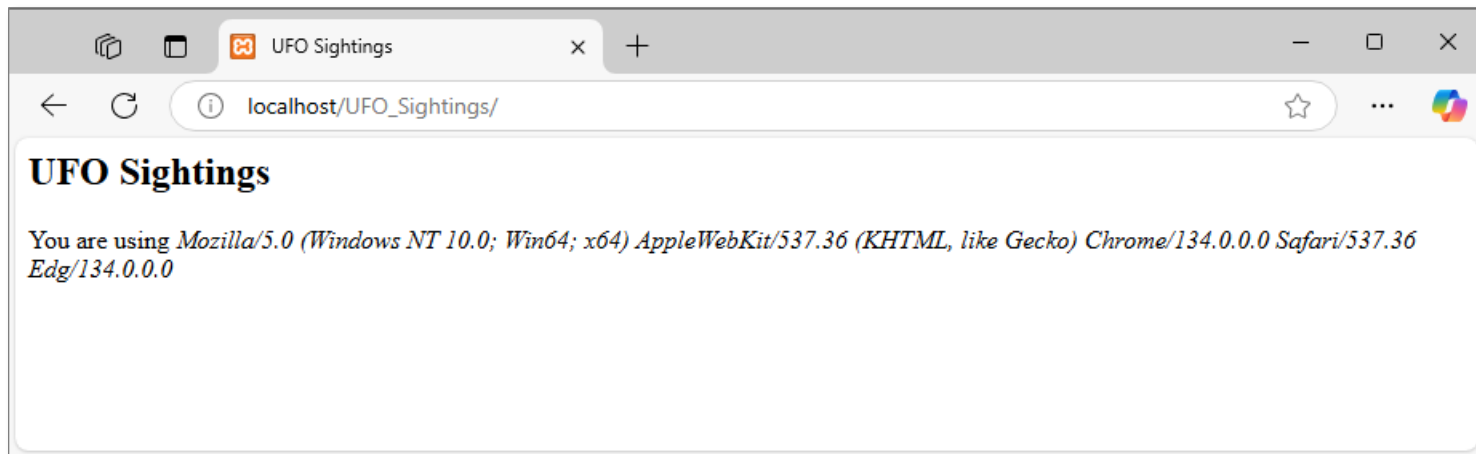


Create index.php file

- Create the **UFO_Sightings/index.php** file

```
<html>
  <head><title>UFO Sightings</title></head>
  <body>
    <h2>UFO Sightings</h2>
    <?php
      $browser = $_SERVER['HTTP_USER_AGENT'];
      echo "<p>You are using <i>$browser</i></p>";
    ?>
  </body>
</html>
```

- Start web server
- Open **http://localhost:????/UFO_Sightings**

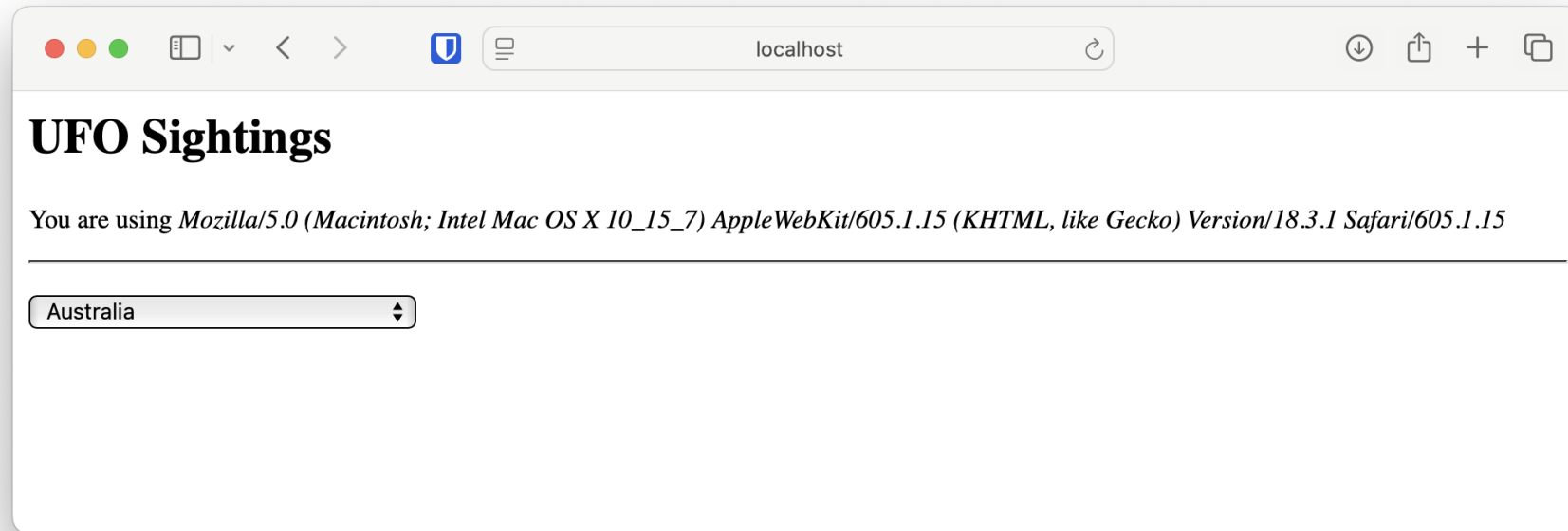


Update index.php file

(...)

```
<hr/>
<select name="Countries" id="Countries">
<?php
    $countries = array(
        "AU" => "Australia",
        "BR" => "Brazil",
        "CA" => "Canada",
        "GB" => "United Kingdom of Great Britain",
        "PT" => "Portugal"
    );
    foreach ($countries as $key => $value) {
        echo "<option value='$key'> $value </option>";
    }
?>
</select>
</body>
</html>
```

Result



- ✓ Australia
- Brazil
- Canada
- United Kingdom of Great Britain
- Portugal

Good Practices

- Create **UFO_Sightings/BusinessLogicLayer.php** file.

```
<?php

function getHttpAgent(){
    $browser = $_SERVER['HTTP_USER_AGENT'];
    echo "<p>You are using <i>$browser</i></p>";
}

function setDropBoxOptions() {
    $countries = array(
        "AU" => "Australia",
        (...)
        "PT" => "Portugal"
    );
    foreach ($countries as $key => $value) {
        echo "<option value='$key'> $value </option>";
    }
}

?>
```


Good Practices

- Update the **UFO_Sightings/index.php** file

```
▶ <?php include "BusinessLogicLayer.php"; ?>

    <html>
        <head><title>UFO Sightings</title></head>
        <body>
            <h2>UFO Sightings</h2>
            ▶ <?php getHttpAgent(); ?>
            <hr/>
            <select name="Countries" id="Countries">
            ▶ <?php setDropBoxOptions(); ?>
            </select>
        </body>
    </html>
```

Get Today's Date

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

```
<?php
```

```
(...)
```

```
function getTodaysDate() {  
    echo "Today is " . Date("d/m/Y") . "<br/>";  
}
```

```
?>
```

- Update the **UFO_Sightings/index.php** file

Send information

- Update the **UFO_Sightings/index.php** file

```
<?php include "BusinessLogicLayer.php"; ?>
<html>
  <head><title>UFO Sightings</title></head>
  <body>
    <div style="text-align:right;width:100%;"><?php getTodaysDate(); ?></div>
    <h2>UFO Sightings</h2>
    <?php getHttpAgent(); ?>
    <hr/>
    <?php if (!isThisACallback()) { ?>
      (...)
    <?php } else { printInfo();} ?>
  </body>
</html>
```

Send information

- Update the **UFO_Sightings/index.php** file

```
(...)  
▶ <?php if (!isThisACallback()) { ?>  
▶     <form action="." method="POST">  
        Date: <input type="date" id="Datetime" name="Datetime"/><br/>  
        City <input type="text" id="City" name="City"/><br/>  
        Country:  
        <select name="Countries" id="Countries">  
            <?php setDropDownOptions(); ?>  
        </select><br/>  
        <p/>  
        <input type="submit" value="Save"/>  
    </form>  
▶ <?php } else { printInfo(); } ?>  
(...)
```

Is a call back?

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

```
<?php
```

```
(...)
```

```
function isThisACallback(): bool{  
    if( empty($_POST['Datetime']) || empty($_POST['City']) || empty($_POST['Countries'])) {  
        return false;  
    }  
    return true;  
}
```

```
(...)
```

```
?>
```

Is a call back! – Print information

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

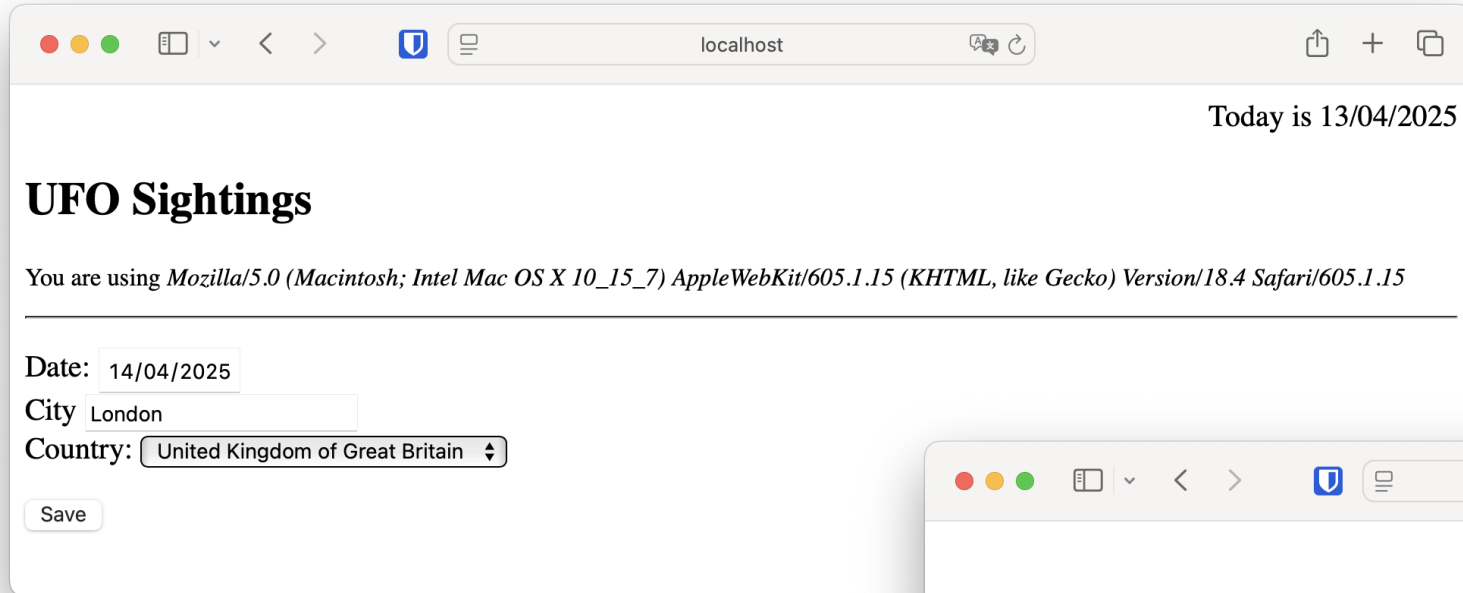
```
<?php
```

```
(...)
```

```
function printInfo() {  
    echo 'Date: ' . $_POST['Datetime'] . '<br/>';  
    echo 'City: ' . htmlspecialchars($_POST['City']) . '<br/>';  
    echo 'Country: ' . $_POST['Countries'] . '<br/>';  
}
```

```
?>
```

Result



A screenshot of a web browser window displaying a form titled "UFO Sightings". The browser's address bar shows "localhost". The page content includes a date "Today is 13/04/2025" in the top right. Below the title, a user agent string is displayed: "You are using Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.4 Safari/605.1.15". The form contains three input fields: "Date:" with the value "14/04/2025", "City" with the value "London", and "Country:" with a dropdown menu showing "United Kingdom of Great Britain". A "Save" button is located at the bottom left of the form.

localhost

Today is 13/04/2025

UFO Sightings

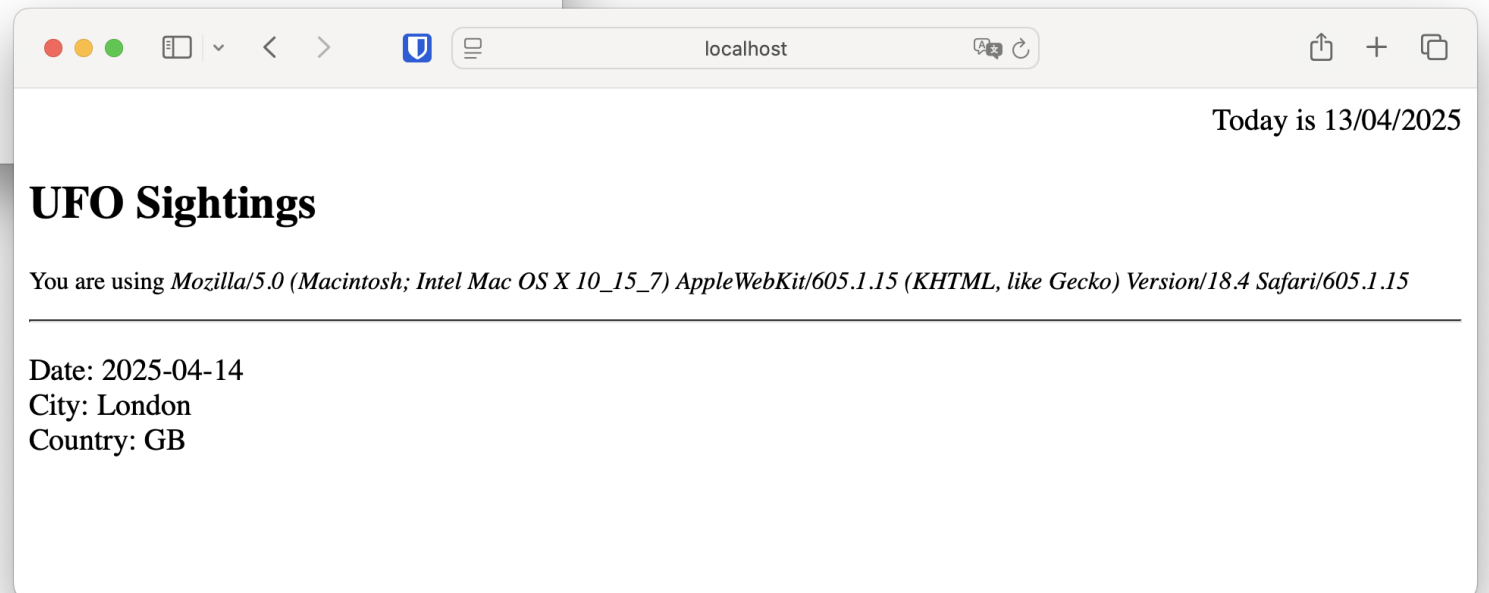
You are using *Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.4 Safari/605.1.15*

Date: 14/04/2025

City London

Country: United Kingdom of Great Britain

Save



A screenshot of a web browser window displaying the result of the "UFO Sightings" form. The browser's address bar shows "localhost". The page content includes a date "Today is 13/04/2025" in the top right. Below the title, the same user agent string is displayed: "You are using Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.4 Safari/605.1.15". The form fields are now populated with the submitted data: "Date: 2025-04-14", "City: London", and "Country: GB".

localhost

Today is 13/04/2025

UFO Sightings

You are using *Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.4 Safari/605.1.15*

Date: 2025-04-14

City: London

Country: GB

UFO Sightings Information

- Information requirements:

Name	Type	Nullable
datetime	datetime	no
city	varchar(70)	yes
country	varchar(6)	no

Is a call back?

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

```
<?php
```

```
(...)
```

```
function isThisACallback(): bool{  
    if(empty($_POST['Datetime']) || empty($_POST['Countries'])){  
        return false;  
    }  
    return true;  
}
```

```
(...)
```

```
?>
```

Is a call back! – Print information

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

```
<?php
```

```
(...)
```

```
function printInfo() {  
    echo 'Date: ' . $_POST['Datetime'] . '<br/>';  
    if(!empty($_POST['City'])){  
        echo 'City: ' . htmlspecialchars($_POST['City']) . '<br/>';  
    }  
    echo 'Country: ' . $_POST['Countries'] . '<br/>';  
}  
  
?>
```

UFO Sightings Information

- Information requirements:

Name	Type	Nullable
datetime	datetime	no
city	varchar(70)	yes
country	varchar(6)	no
shape	varchar(30)	yes
duration	int	no
comments	text	no
latitude	double	no
longitude	double	no

Result

localhost

Today is 15/04/2025

UFO Sightings

You are using *Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Safari/605.1.15*

Date: 15/04/2025

City: Porto

Country: Portugal

Shape: Diamond

Duration: 37

Comments:

It appeared over the sea near the mouth of the Ouro River. It stabilized about 400 meters above the sea, rocked for 10 seconds and then accelerated vertically upwards.

Latitude: 41.143251

Longitude: -8.677609

Save

localhost

Today is 15/04/2025

UFO Sightings

You are using *Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.4 Safari/605.1.15*

Date: 2025-04-15

City: Porto

Country: PT

Shape: Diamond

Duration: 37

Comments: It appeared over the sea near the mouth of the Ouro River. It stabilized about 400 meters above the sea, rocked for 10 seconds and then accelerated vertically upwards.

Latitude: 41.143251

Longitude: -8.677609

Good Practices

- Update the **UFO_Sightings/index.php** file

```
<?php include "BusinessLogicLayer.php"; ?>
<html>
  <head><title>UFO Sightings</title></head>
  <body>
    <div style="text-align:right;width:100%;"><?php getTodaysDate() ?></div>
    <h2>UFO Sightings</h2>
    <?php getHttpAgent(); ?>
    <hr/>
    <p>
      <?php showUI(); ?>
    </p>
  </body>
</html>
```

Good Practices

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

```
<?php
(...)
function displayForm() {
    echo '<form action="." method="POST">
        Date: <input type="date" id="Datetime" name="Datetime"/><br/>
        City: <input type="text" id="City" name="City"/><br/>
        Country: <select name="Countries" id="Countries">';
    setDropBoxOptions();
    echo ' </select><br/>
        Shape: <input type="text" id="Shape" name="Shape"/><br/>
        Duration: <input type="number" id="Duration" name="Duration" min="1" step="1"/><br/>
        Comments: <textarea name="Comments" id="Comments" rows="4" cols="50"></textarea><br/>
        Latitude: <input type="number" id="Latitude" name="Latitude" step="any"/><br/>
        Longitude: <input type="number" id="Longitude" name="Longitude" step="any"/><br/>
        <p></p>
        <input type="submit" value="Save"/>
        </form>';
}
?>
```

Good Practices

- Update **UFO_Sightings/BusinessLogicLayer.php** file.

```
<?php

(...)

function showUI(){
    if(!isThisACallback()){
        displayForm();
    }
    else{
        printInfo();
    }
}
?>
```